

COURSE 3 - DATA ENGINEERING 2026**C3-06 - Targeted Remediation Cards**

Repair only the weak streaming competency, then prove it on a changed micro-case.

Artifact	C3-06-REPAIR
Status	2026 - LEARNER EDITION
Phase	C3-06 Streaming & Event-Driven Pipelines 13 lessons 34-46 focused hours
Prerequisite	C3-05 Gate PASS; C2B streaming/lakehouse concepts may be recalled but Kafka/Spark streaming implementation is taught here.
Current lane	Apache Kafka 4.3.1 Spark Structured Streaming 4.2.0 Delta Lake 4.4.0 Debezium 3.6.2.Final
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny event example -> guided Kafka/streaming action -> expected result -> why it worked -> break/fix -> independent changed case -> validation -> replay/state/production reasoning -> evidence. A running stream is not mastery if duplicates, late data, offsets, checkpoints or sink state are wrong.

Repair routing

Architecture/event contract

Re-state event identity, event time, latency/replay/state requirement and choose processing mode again on a changed workload.

Kafka key/offset/replay

Draw partitions and offsets for a different entity key, then explain replay/retention and group behavior.

Schema/CDC

Evaluate a changed schema version and one insert/update/delete CDC envelope without copying the Gate solution.

Duplicates/idempotency

Process the same event twice against a fresh target state and prove one business effect.

Event time/watermark

Use a different late-event timeline/window size; predict state/late handling before running.

Checkpoint/restart

Compare same-checkpoint restart with fresh checkpoint on a changed source; no checkpoint surgery.

Lakehouse sink

Redesign a fresh Bronze/Silver sink with separate checkpoints and an idempotent write boundary.

Observability/runbook

Diagnose a new source-stall/backlog/state-growth snapshot and write detect -> diagnose -> recover -> validate steps.

A repaired micro-case does not automatically assign Gate B/C. Fresh full retest assignment is explicit and remains sealed until then.